Research and caveats summary

DEMO-EH-1004 · *Boswellia papyrifera* · resin (stem tapping) · status: Quality review · **SYNTHETIC DEMO DATA**

Activity: frankincense tapping · Region: Tigray / Amhara lowlands, Ethiopia (synthetic) · Jurisdiction: Ethiopia · Exact site: —

A Harvest Passport is an internal evidence and traceability record. It is not a government permit, a FairWild certificate, a validated local harvest quota, or an assurance of legal compliance or product safety.

Species- and part-specific cautions

Status: ok

Published observational comparison (harvest not randomised) on resin (stem tapping) harvest (supported-by-study; applies: same species, same part; review: awaiting expert review)

Mean long-term growth rate λ 0.962 in harvested populations vs 0.954 in the comparison populations (difference +0.008; 1 vs 1 matrices).

Next: Use it as a scenario in the Studio; local stage counts and recovery measurements are still needed before setting any limit.

Sources: Groenendijk P, Eshete A, Sterck FJ, Zuidema PA, Bongers F (2012) Limitations to sustainable frankincense production: blocked regeneration, high adult mortality and declining populations. Journal of Applied Ecology 49:164-173. doi:10.1111/j.1365-2664.2011.02078.x (https://doi.org/10.1111/j.1365-2664.2011.02078.x)

How sensitive this population is to the harvested process (reproduction) (computed-from-matrices; applies: this species; review: needs expert review before use)

About 1% (95% range 1–1%) of this plant's long-run population growth depends on reproduction, the process a resin harvest affects (2 published matrices, this species).

Next: Use this to judge which local measurements matter most (e.g. survival of large plants), not as a harvest limit.

Sources: COMPADRE v6.26.3.0 matrices (computed here)

1 paper on this species reports that harvest increased growth. (ai-found-literature; applies: this species; review: needs expert review before use)

Found by the evidence agent: JEV classified the paper, Groq extracted the finding, and the quote below was checked word-for-word against the abstract.

Next: A scientific reviewer should read the paper(s) and confirm or reject this finding.

Sources: Limitations to sustainable frankincense production: blocked regeneration, high adult mortality and declining populations (https://doi.org/10.1111/j.1365-2664.2011.02078.x)

Federal forest law applies (regulatory-listing; applies: Ethiopia; review: awaiting expert review)

Collection in forests falls under the Forest Development, Conservation and Utilization Proclamation No. 1065/2018. We did not locate a provision specific to frankincense tapping.

Next: Confirm tapping rights, concession or community forest agreements with the regional forest authority.

Sources: Forest Development, Conservation and Utilization Proclamation No. 1065/2018 (Federal Negarit Gazette, Ethiopia) (https://www.metaappz.com/References/ethiopian_laws/federal/pr_1065_2018/en/pdf)

4 papers the AI could not classify confidently (needs-review; applies: search results; review: awaiting expert review)

JEV and the keyword rules disagreed, or JEV was unsure. They may or may not contain harvest effects.

Next: A scientific reviewer should screen these and promote relevant findings.

Sources: Climate change and the sustainable use of medicinal plants: a call for "new" research strategies. (2024) (https://europepmc.org/article/MED/39963365); Composition, medicinal values, and threats of plants used in indigenous medicine in Jawi District, Ethiopia: implications for conservation and sustainable use. (2024) (https://europepmc.org/article/MED/39384598); Ethiopian vegetation types, climate and topography. (2020) (https://europepmc.org/article/MED/33094201); Ethnobotanical study of traditional medicinal plants used by the local Gamo people in Boreda Abaya District, Gamo Zone, southern Ethiopia. (2024) (https://europepmc.org/article/MED/38419092)

Legal compliance is part of sustainable collection (general; applies: any wild collection; review: awaiting expert review)

Voluntary standards treat legal compliance at local, national and international level as a precondition.

Next: List the authorisations your collection needs and attach them to the passport.

Sources: FairWild Standard: Version 3.0 (approved 19 December 2023) (https://www.fairwild.org/s/FairWildStandard-V3-FINAL.pdf)

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Customary rights and benefit sharing (general; applies: any wild collection; review: awaiting expert review)

Collection should respect collectors' and communities' customary rights and share benefits fairly.

Next: Record how consent and benefit sharing were agreed with the community that holds rights to the site.

Sources: FairWild Standard: Version 3.0 (approved 19 December 2023) (https://www.fairwild.org/s/FairWildStandard-V3-FINAL.pdf)

Collection and post-harvest quality practice (general; applies: any wild collection; review: awaiting expert review)

GACP guidance covers identification, collection practice, drying, storage and documentation of herbal starting materials.

Next: Keep identification, drying and storage records per lot; ask the buyer which GACP expectations apply.

Sources: WHO guidelines on good agricultural and collection practices (GACP) for medicinal plants (2003) (https://www.who.int/publications/i/item/9241546271); EMA Guideline on Good Agricultural and Collection Practice (GACP) for starting materials of herbal origin (EMEA/HMPC/246816/2005; Revision 1 adopted by HMPC in 2025) (https://www.ema.europa.eu/system/files/documents/scientific-guideline/hmpc-guideline-gacp-may-2025_en.pdf)

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